A taxonomy of service identification approaches for legacy software systems modernization

Citation: M. Abdellatif *et al.* "A taxonomy of service identification approaches for legacy software systems modernization," *Journal of Systems and Software*, 2021. [Online]. Available: https://linkinghub.elsevier.com/retrieve/pii/S0164121220302582

Paper Information: Systematic Literature Review (SLR)

Rating: ⭐ ⭐ ⭐ ⭐

Own Keywords: Background, Analysis of Service Identification Approaches (SIA), Migration, SIA Taxonomy

Reading Progress: Second Pass

Introduction (What is the Paper about?)

- Paper **performed an SLR** to **create a taxonomy** to categorize *Service Identification Approaches (SIA)* focusing on *bottom-up* and *hybrid strategies*
- Paper aims to study SIAs in literature along four dimensions: *Used Inputs*, *Applied Processes*, *Resulting Outputs* and *Usability Degree*
- **Goal: Help Practitioners in selecting a suitable SIA**

Methods

- SLR covering 41 published SIAs (retained from a first set of 3,246 Papers) based on software-systems analyses
  - **build SLR based on own experience** with migration and experiences working in industry
  - focusing on *bottom-up* and *hybrid* approaches that rely on existing software artifacts
  - **Reason:** Studies and experience have shown that companies often only have source code as most up-to-date source of information about their legacy systems
    - *(Comment)* Missing requirement engineering data and artifacts during the software designing process are hindering a *top-up* approach ?
- **Validation of correctness and coverage** through a **survey** and **one-to-one interviews** with 45 **industrial experts** who are performing migration

Scientific Benefit

- **Taxonomy to categorize *Service Identification Approaches (SIA)***
  - Build a multi-layer classification starting from high-level, based on *Used Inputs*, *Applied Processes*, *Given Outputs* and the *SIAs Usability*
  -
- **Most existing SIAs look for services based on functional cohesion and low coupling towards other parts of the applications**
  - disregard service types
  - current trend is to identify microservices by applying the same techniques used for identifying "normal" services
- **Results show that classification suits industrial experts' experience**
  - Precision of 99% and a recall of 94%

Challenges / Future Work

- **Most studied SIAs are still at their infancy**
  - *(Comment)* May be potentially outdated
  - lack of validation on real enterprise-scale systems
  - lack of tool support
  - lack of automation of SIAs
  - lack of quality assessment of the identified services
- **SIAs generally ignore economic aspects** like *implementation*, *maintenance*, *re-factoring costs and time-to-market issues*
  - only a few SIAs consider economic aspects
- **SIAs should provide means to assess their quality and consider economic aspects**
- **Authors are convinced that more work should be done to automate** state-of-the-art SIAs
- **Granularity between services and microservices is still not well defined** nor clearly discussed by SIA approaches

Limitations

Paper Outline and Read Progress

1. [X] Introduction
2. [X] SLR Methodology
3. [X] Inputs used by SIAs
4. [ ] Applied Processes
5. [ ] Outputs generated by SIAs
6. [ ] Usability Level of these SIAs
7. [ ] Validation of the taxonomy
8. [X] Comparison between studied SIAs
9. [X] Related Work
10. [X] Conclusion

Own Comments

- (+) Key Paper
- (-) No special focus on microservices
- (-) Potential subjective because the authors mentioned they build their SLR based on their own experience